Azure Data Engineer
SQL Deep-Dive Interview Questions

Scenarios · Edge Cases · Performance · Azure Synapse · Databricks

Questions Only — No Answers Provided

Practice independently and build deep expertise

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Section 1 — SQL Fundamentals & Query Mechanics

Q1. Explain the logical order of SQL clause execution. How does this affect the use of column aliases in WHERE vs HAVING?

Q2. What is the difference between WHERE and HAVING? Can you use aggregate functions in a WHERE clause? Justify your answer with an example.

Q3. You have a table with 10 million rows. A query using SELECT DISTINCT on a non-indexed column is running slowly. What are the possible causes and how would you optimise it?

■ *Focus: Performance & Indexing*

Q4. Describe the difference between UNION and UNION ALL. In what scenarios would UNION ALL be preferred in an Azure Synapse dedicated pool?

Q5. What is a correlated subquery? Write an example that retrieves the latest order per customer from an Orders table without using window functions.

Q6. Explain the difference between EXISTS and IN. When can EXISTS outperform IN, and when does IN perform better?

Q7. You need to return the second-highest salary from an Employees table without using TOP, LIMIT, or window functions. How would you do it?

Q8. What are the implications of using SELECT * in production ETL pipelines, particularly in Azure Data Factory or Synapse pipelines?

Q9. Explain the difference between CHAR, VARCHAR, NVARCHAR, and VARCHAR(MAX) in SQL Server / Synapse. When would you choose each?

■ *Focus: Data types & storage*

Q10. A query returns duplicate rows even after applying DISTINCT. What are the possible root causes and how would you debug it?

Q11. What is a derived table? Compare it to a CTE and explain when you would choose one over the other.

Q12. Describe implicit vs explicit type casting in T-SQL. Provide a scenario where implicit casting causes a performance regression.

Q13. How does T-SQL handle division by zero? What are the safe patterns to avoid runtime errors in production pipelines?

Q14. You have a query that filters on a computed column in the WHERE clause. The query is doing a full table scan. Explain why and how to fix it.

Q15. What is the APPLY operator (CROSS APPLY / OUTER APPLY)? Give a real-world scenario where it is the best tool for the job.

Section 2 — Window Functions & Analytics

Q16. Explain the difference between ROW_NUMBER(), RANK(), and DENSE_RANK(). Give a concrete example where each produces a different result.

Q17. How do LEAD() and LAG() work? Write a query to calculate the day-over-day percentage change in sales using a Sales table with columns (date, revenue).

Q18. What is the difference between ROWS BETWEEN and RANGE BETWEEN in a window frame? Give an example where they produce different results.

Q19. You need to compute a 7-day rolling average of daily active users. Write the SQL and explain the window frame specification.

Q20. Using window functions only, write a query to identify gaps and islands in a sequence of consecutive dates per user_id.

Q21. How does PARTITION BY affect the behaviour of window functions? What happens when PARTITION BY is omitted?

Q22. What is CUME_DIST() and PERCENT_RANK()? Describe a business scenario — such as a sales leaderboard — where you'd use each.

Q23. A data analyst asks you to flag the top 10% of customers by lifetime spend within each region. Write the SQL using window functions.

Q24. Explain NTILE(n). A product manager wants to split customers into quartiles based on order frequency. How would you implement this?

Q25. How would you use window functions to detect duplicate records in a staging table without deleting them immediately?

Q26. FIRST_VALUE() and LAST_VALUE() are window functions. Why does LAST_VALUE() often return unexpected results, and how do you fix it?

Q27. You have session-level clickstream data. Write a query using window functions to compute the time spent in each step of a user funnel.

Section 3 — CTEs, Subqueries & Recursive SQL

Q28. What is a Common Table Expression (CTE)? What are the advantages of CTEs over subqueries in terms of readability and performance?

Q29. Are CTEs materialised in SQL Server and Azure Synapse Dedicated Pool? How does this affect performance compared to temp tables?

Q30. Write a recursive CTE to traverse an employee organisational hierarchy starting from a given manager_id and returning all direct and indirect reports.

Q31. A recursive CTE is causing infinite loops in production. What are the causes and safeguards you would put in place?

Q32. How would you use a recursive CTE to generate a date dimension table covering all dates between two given dates?

Q33. Explain when you would use multiple CTEs chained together vs a single complex subquery. What are the trade-offs?

Q34. Write a query to find all products that have never been ordered, using a subquery, then rewrite it using a CTE, and finally using a LEFT JOIN. Compare the three approaches.

Q35. You have a deeply nested JSON column. How would you combine CTEs with OPENJSON or JSON_VALUE in Synapse to flatten hierarchical data?

■ *Focus: JSON in T-SQL*

Q36. What is the maximum recursion depth for recursive CTEs in SQL Server / Synapse? How would you handle scenarios requiring deeper traversal?

Q37. Describe a scenario in an Azure Data Warehouse where replacing a correlated subquery with a CTE dramatically improved query performance.

Section 4 — Joins, Set Operations & NULL Handling

Q38. Explain the difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, and CROSS JOIN with real-world use cases for each.

Q39. What is a self-join? Write a query to find all pairs of employees who share the same manager.

Q40. What are the differences between INTERSECT, EXCEPT, and UNION in terms of behaviour with NULLs and duplicates?

Q41. In a LEFT JOIN, you add a WHERE clause filtering on a column from the right table. What result does this produce and why? How do you fix it to preserve left-side rows?

Q42. You join two large tables in Synapse and the query is extremely slow. Walk through your diagnostic and optimisation steps.

■ *Focus: Synapse performance*

Q43. What is a HASH JOIN vs a MERGE JOIN vs a NESTED LOOP JOIN in query execution plans? When does the optimiser choose each?

Q44. Explain three-valued logic in SQL. How does NULL affect the result of comparisons, and what are the implications for JOIN conditions?

Q45. Write a query that returns rows where a nullable column contains NULL values. What common mistakes do developers make here?

Q46. What is COALESCE and how does it differ from ISNULL in T-SQL? Are there performance differences?

Q47. You have two tables with millions of rows. Both have a common key, but the key in one table is VARCHAR and in the other it is INT. What happens during the join and how do you fix it?

Q48. Explain NULLIF(). Provide a business scenario where NULLIF() prevents a divide-by-zero error in a KPI calculation.

Q49. What is a non-equi join? Give a practical example — such as matching transactions to date ranges or pricing tiers.

Section 5 — Indexing, Partitioning & Performance Tuning

Q50. What is a clustered index vs a non-clustered index? Can a table have both? What are the trade-offs?

Q51. Explain index selectivity. Why does an index on a low-cardinality column (e.g., gender) often not get used by the optimiser?

Q52. What is a covering index? Write a scenario where adding an INCLUDE column to a non-clustered index eliminates key lookups.

Q53. What is index fragmentation and how does it impact query performance? How do you detect and resolve it in Azure SQL / Synapse?

Q54. Explain table partitioning in SQL Server and Synapse. What are the key benefits and limitations for large fact tables?

Q55. What is partition elimination? Write an example query on a date-partitioned table and explain how the optimiser prunes partitions.

Q56. A query that was fast last month is now slow. The data volume hasn't changed significantly. Walk through a systematic troubleshooting approach.

■ *Focus: Query regression*

Q57. What are statistics in SQL Server / Synapse and why are they important? How do you detect stale statistics and update them?

Q58. Explain the difference between a table scan, index seek, and index scan in an execution plan. Which is best and in what context?

Q59. What is a parameter sniffing problem in T-SQL? How does it manifest and what are the remediation strategies?

Q60. In Azure Synapse Dedicated Pool, explain the difference between Round-Robin, Hash, and Replicated distribution strategies. How do you choose the right one?

■ *Focus: Synapse distribution*

Q61. What are columnstore indexes? How do they differ from rowstore indexes, and when should you use them in a data warehouse?

Q62. A Synapse query has a high number of data movement operations (DMS). What does this mean and how do you reduce it?

■ *Focus: Synapse MPP*

Q63. What is the NOLOCK hint? What are the risks of using it in a production ETL pipeline?

Section 6 — Transactions, Isolation Levels & Concurrency

Q64. Explain ACID properties. How does Azure Synapse Dedicated Pool differ from Azure SQL Database in its ACID support?

Q65. What are the four SQL transaction isolation levels? Describe the phenomena (dirty read, non-repeatable read, phantom read) each prevents.

Q66. What is a deadlock? Walk through a real-world scenario involving two tables and two sessions. How do you detect and prevent it?

Q67. What is optimistic vs pessimistic concurrency control? How does Snapshot Isolation differ from Serializable isolation?

Q68. You are running a long ETL transaction that locks a critical table. How would you redesign the pipeline to minimise lock duration?

Q69. Explain the difference between explicit transactions (BEGIN TRAN / COMMIT / ROLLBACK) and implicit transactions in SQL Server.

Q70. What is a savepoint in a transaction? Provide a use case where a partial rollback is needed in a multi-step ETL process.

Q71. What is READ_COMMITTED_SNAPSHOT (RCSI)? How does it improve concurrency and what are its trade-offs?

Q72. In Azure Synapse Dedicated Pool, what level of transaction support is available? How does this impact multi-step ETL design?

Q73. Describe how you would implement an idempotent MERGE operation to safely re-run a pipeline without producing duplicate data.

Section 7 — Azure Synapse Analytics (Dedicated SQL Pool)

Q74. What is Massively Parallel Processing (MPP) in Azure Synapse? How does data distribution affect query parallelism?

Q75. You need to load 500 GB of Parquet files from ADLS Gen2 into a Synapse Dedicated Pool daily. Describe your end-to-end loading strategy.

■ *Focus: PolyBase / COPY*

Q76. What is PolyBase and how does it differ from the COPY INTO statement in Synapse? When would you use each?

Q77. Explain the concept of data skew in Synapse. How do you detect skew and what are the remediation strategies?

Q78. What is a replicated table in Synapse? When is it appropriate and what are the maintenance implications after DML operations?

Q79. Describe the Synapse resource class system. How do resource classes affect query concurrency, memory, and performance?

Q80. What is workload management in Synapse (workload groups and classifiers)? How would you configure it for mixed ETL and reporting workloads?

Q81. A Synapse query plan shows a large BroadcastMoveOperation. What causes this and how do you eliminate it?

Q82. Explain the CTAS (CREATE TABLE AS SELECT) pattern. What are its advantages over INSERT INTO SELECT in Synapse?

Q83. How would you implement slowly changing dimensions (SCD Type 2) efficiently in a Synapse Dedicated Pool?

Q84. What are result set caching and query plan caching in Synapse? How do you enable, validate, and invalidate the result set cache?

Q85. Describe the STATISTICS object in Synapse. How is auto-create statistics different from manually created statistics for MPP workloads?

Q86. You are designing a large fact table in Synapse. Walk through your decisions on distribution key, partition column, and index type.

Q87. How do you handle schema evolution (adding/changing columns) in a Synapse Dedicated Pool without downtime?

Q88. What is the EXPLAIN statement in Synapse / Parallel Data Warehouse? How do you interpret its output to optimise queries?

Section 8 — Azure Synapse Serverless SQL Pool

Q89. What is Synapse Serverless SQL Pool and how does it differ architecturally from the Dedicated Pool? When would you choose each?

Q90. Write a query using OPENROWSET to read a Parquet file from ADLS Gen2. What are the key parameters and security considerations?

Q91. What are external tables and views in Serverless SQL Pool? How do you create and manage them for a Delta Lake layer?

Q92. How does Serverless SQL Pool handle schema inference for Parquet, CSV, and JSON? What are the limitations?

Q93. Describe how you would expose a Serverless SQL external table as a data source for Power BI. What are the performance trade-offs?

Q94. A Serverless SQL query on a large Parquet dataset is scanning too many files. How do you implement partition pruning and file pruning?

Q95. What is the Delta format, and how does Serverless SQL Pool support reading Delta tables including time travel queries?

Q96. How do you implement row-level security and column masking in Synapse Serverless SQL for different consumer roles?

Q97. Explain the cost model for Synapse Serverless SQL. What query patterns are most expensive and how do you control costs?

Q98. What are the limitations of DML operations in Serverless SQL Pool, and how do you work around them for data updates?

Section 9 — Delta Lake & Databricks SQL

Q99. What is Delta Lake? Explain how it adds ACID transactions on top of a data lake. How does the transaction log work?

Q100. Explain the difference between Delta Lake, Apache Iceberg, and Apache Hudi. When would you choose Delta on Azure?

Q101. What is the MERGE INTO (UPSERT) statement in Delta Lake? Write a query to implement an SCD Type 1 merge pattern.

Q102. What is Delta Lake time travel? Write a SQL query to read a Delta table as of a specific timestamp and as of a specific version.

Q103. What is Z-Ordering in Delta Lake? How does it improve query performance, and what are its limitations?

Q104. What is OPTIMIZE in Delta Lake? How does it relate to the small files problem and how do you schedule it in Databricks?

Q105. What is VACUUM in Delta Lake? What happens to time travel capabilities after running VACUUM? What is the default retention period?

Q106. Describe schema enforcement vs schema evolution in Delta Lake. How do you enable schema evolution on a MERGE or write operation?

Q107. How does Change Data Feed (CDF) work in Delta Lake? How would you use it to implement incremental pipelines?

Q108. You are running a streaming write and a batch read simultaneously on the same Delta table. How does Delta Lake handle concurrent writes?

Q109. What is Delta Live Tables (DLT)? How does it differ from a standard Databricks notebook-based pipeline?

Q110. Describe how you would implement a Medallion Architecture (Bronze/Silver/Gold) using Delta Lake on Azure Data Lake Storage Gen2.

Section 10 — Data Modelling & Data Warehouse Patterns

Q111. Compare Star Schema and Snowflake Schema. What are the query performance implications of each in Azure Synapse?

Q112. What is a degenerate dimension? Give an example and explain when you would use it.

Q113. Explain SCD Type 0, 1, 2, 3, 4, and 6. For a customer address change, which type is most appropriate and why?

Q114. What is a fact-less fact table? Describe two real-world use cases where it makes sense.

Q115. What is the Medallion Architecture and how does it map to traditional Data Warehouse layers (staging, ODS, DWH, Data Mart)?

Q116. How do you handle late-arriving facts and late-arriving dimensions in a data warehouse pipeline?

Q117. What is a conformed dimension? Why is it critical for cross-subject-area reporting?

Q118. Describe surrogate keys vs natural keys in a data warehouse. What are the risks of using natural keys as fact table foreign keys?

Q119. What is a bridge table? Describe a scenario involving a many-to-many relationship — such as employees and skills — where one is required.

Q120. How do you implement data vault modelling (Hub, Link, Satellite)? How does it compare to Kimball in an Azure context?

Q121. You are asked to design the schema for a near-real-time IoT sensor analytics system on Azure Synapse. Walk through your design decisions.

Q122. What is a mini-dimension and when would you use it to optimise a rapidly changing large dimension?

Section 11 — Real-World Scenario Questions

Q123. Scenario: Your production Synapse pipeline fails nightly with a timeout. Investigation shows one query has gone from 30 seconds to 45 minutes over 3 months. The table has grown from 50M to 800M rows. There are no code changes. Describe your full diagnostic and remediation approach.

■ *Focus: Performance regression at scale*

Q124. Scenario: A business analyst reports that total revenue figures in the data warehouse differ by 2% from the source ERP system. Walk through your end-to-end data reconciliation process.

■ *Focus: Data quality & reconciliation*

Q125. Scenario: You need to ingest 200 source tables from an on-premises SQL Server into Azure Data Lake Gen2 as Delta tables, maintaining daily incremental loads using watermark columns. Design the architecture and write the key SQL patterns.

■ *Focus: Incremental ingestion architecture*

Q126. Scenario: A finance team needs a self-service SQL view that always shows each salesperson's YTD revenue, rank within their region, and percentage contribution to regional total — refreshed hourly. Design the underlying table structure, the SQL view, and the refresh strategy.

■ *Focus: Aggregation & window functions*

Q127. Scenario: You discover that a critical dimension table in Synapse is hash-distributed on a low-cardinality column (e.g., country_code with 5 values), causing severe data skew. How do you identify the skew, choose a better distribution key, and migrate without downtime?

■ *Focus: Distribution strategy redesign*

Q128. Scenario: A Databricks job reads a Delta table shared with a Synapse Serverless pool. After a Delta OPTIMIZE + VACUUM, the Synapse queries start failing with 'file not found' errors. Explain what happened and how you prevent it.

■ *Focus: Delta Lake & Synapse integration*

Q129. Scenario: You are building a GDPR-compliant data platform. A user requests to be forgotten. Describe how you would implement the right-to-erasure across Bronze/Silver/Gold Delta layers, Synapse Dedicated Pool, and downstream exports.

■ *Focus: GDPR & data governance*

Q130. Scenario: Every morning at 9 AM, all BI dashboards in Power BI slow down significantly. The Synapse Dedicated Pool is at DWU6000. What are the likely causes and how do you resolve them using workload management?

■ *Focus: Workload management*

Q131. Scenario: You need to implement a real-time fraud detection feed. Transactions come from Event Hub into Databricks Structured Streaming. The fraud rules are maintained in a Delta lookup table updated every 5 minutes. Describe how you join the stream with the lookup and write flagged transactions to a Gold Delta table.

■ *Focus: Streaming + Delta Lake*

Q132. Scenario: A MERGE statement in Synapse Dedicated Pool is causing deadlocks under concurrent load. Walk through the root cause analysis and redesign the pattern to be deadlock-free.

■ *Focus: Concurrency & locking*

Q133. Scenario: You are asked to compute the retention cohort analysis: for each month, what percentage of customers who first purchased in that month made a second purchase within 30 days, 60 days, 90 days, and 180 days. Write the SQL.

■ *Focus: Cohort analysis*

Q134. Scenario: A partner team is exporting 10 years of transaction history to a CSV for an external audit. The export query is timing out after 2 hours. How would you redesign the extraction approach using Synapse and ADLS Gen2?

■ *Focus: Large export pattern*

Q135. Scenario: You are handed a legacy stored procedure with 3,000 lines of T-SQL that joins 18 tables and takes 4 hours to run. Describe your methodology for analysing, refactoring, and optimising it.

■ *Focus: Query refactoring*

Q136. Scenario: An external vendor delivers daily CSV files with inconsistent date formats, extra columns appearing randomly, and occasional duplicate primary keys. Design a robust ingestion pattern in Synapse / Databricks that handles all these issues and alerts on anomalies.

■ *Focus: Resilient ingestion & data quality*

This document contains 136 questions across 11 categories. All questions are designed for senior / lead Azure Data Engineer interview preparation. No answers are provided — work through each independently to maximise learning.